

■ 1.7.4 Finding the Structure of a List

<code>Length[list]</code>	the number of elements in <i>list</i>
<code>Dimensions[list]</code>	the dimensions of a list that forms an array
<code>VectorQ[list]</code>	test whether <i>list</i> has the structure of a vector
<code>MatrixQ[list]</code>	test whether <i>list</i> has the structure of a matrix
<code>TensorRank[list]</code>	find the rank of <i>list</i> , when viewed as a tensor
<code>MemberQ[list, form]</code>	test whether <i>form</i> is an element of <i>list</i>
<code>Count[list, form]</code>	the number of times <i>form</i> appears as an element of <i>list</i>
<code>FreeQ[list, form]</code>	test whether <i>form</i> occurs nowhere in <i>list</i>
<code>Position[list, form]</code>	the positions at which <i>form</i> occurs in <i>list</i>

Functions for finding the structure of lists.

`Length` gives the number of elements in a list.

```
In[1]:= Length[{a,b,c,d}]
Out[1]= 4
```

The list {} has zero elements.

```
In[2]:= Length[{}]
Out[2]= 0
```

This shows that *a* is an element of {*a*, *b*, *c*}.

```
In[3]:= MemberQ[{a, b, c}, a]
Out[3]= True
```

On the other hand, *d* is not.

```
In[4]:= MemberQ[{a, b, c}, d]
Out[4]= False
```

This counts the number of times that *a* appears as an element of the list.

```
In[5]:= Count[{a, a, a+1, b, a}, a]
Out[5]= 3
```

This assigns *m* to be the 3×3 identity matrix.

```
In[6]:= m = IdentityMatrix[3]
Out[6]= {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}}
```

Here are the dimensions of *m*.

```
In[7]:= Dimensions[m]
Out[7]= {3, 3}
```

This shows that 0 does occur *somewhere* in `m`.
`In[8]:= FreeQ[m, 0]`
`Out[8]= False`

This gives a list of the positions at which 0 occurs in `m`.
`In[9]:= Position[m, 0]`
`Out[9]= {{1, 2}, {1, 3}, {2, 1}, {2, 3}, {3, 1}, {3, 2}}`

The functions `Count` and `Position`, as well as `MemberQ` and `FreeQ`, can in fact be used not only to search for particular list elements, but also for classes of elements which match particular “patterns”. (`Cases` gives the actual elements that occur.) Section 2.3.5 will discuss how this works.